

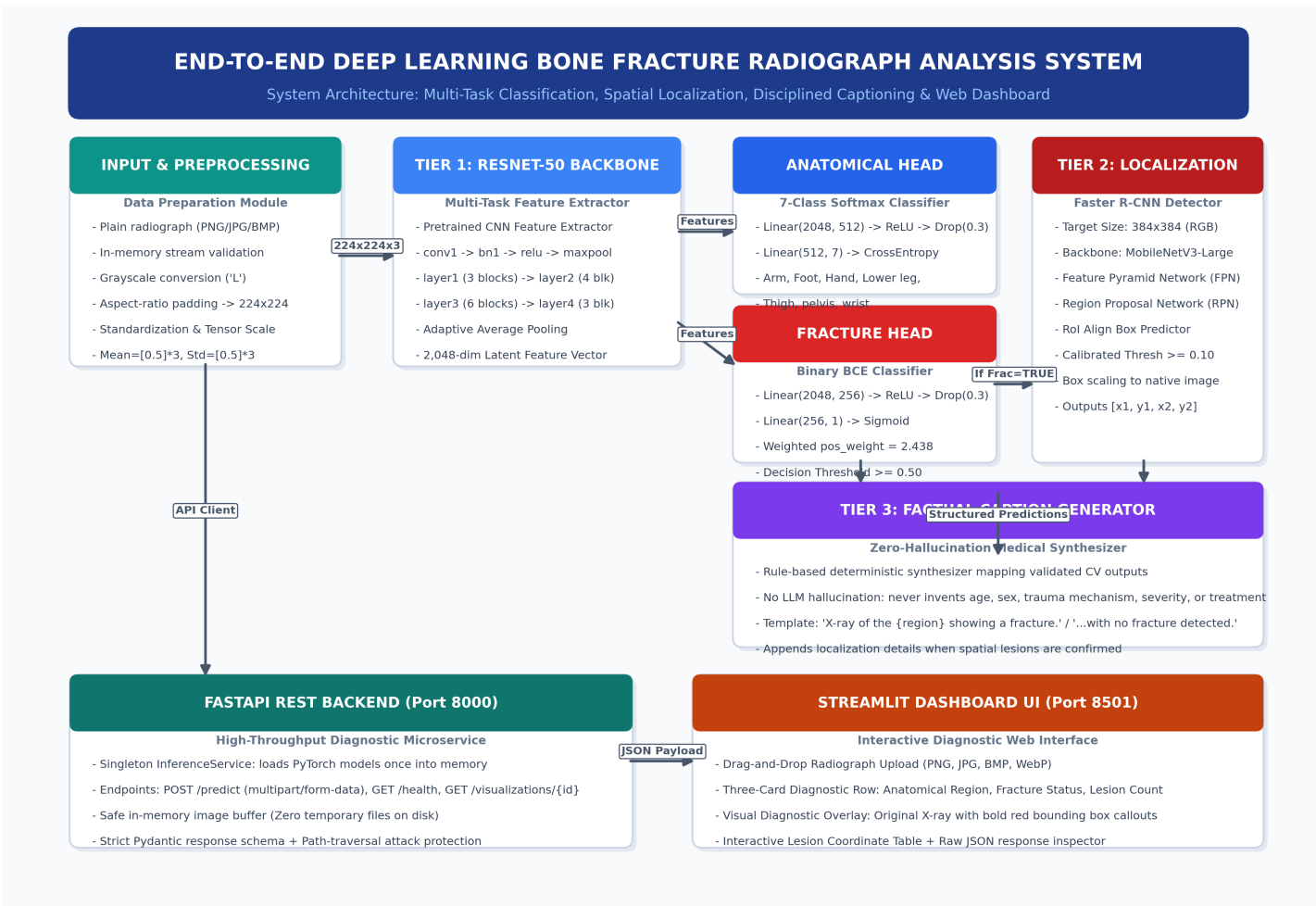
# Bone Fracture Detection & Captioning System

System Architecture Specification & Technical Block Diagram Document | Version 1.0

## 1. System Architectural Overview

The Bone Fracture Image Captioning and Detection platform provides automated, multi-task radiograph analysis consisting of three primary execution tiers: (1) Multi-Task Shared Convolutional Backbone for 7-class anatomical site classification and binary fracture detection, (2) Faster R-CNN with Feature Pyramid Network (FPN) for spatial lesion localization, and (3) Disciplined Factual Caption Synthesizer generating non-hallucinatory medical text.

## 2. System Architecture Block Diagram



## 3. Deep Learning Component Specifications

| Component           | Architecture            | Loss / Objective                 | Output Specification                                        |
|---------------------|-------------------------|----------------------------------|-------------------------------------------------------------|
| Classifier Backbone | ResNet-50 (Shared)      | Two-stage Transfer Learning      | 2,048-dim feature representation                            |
| Anatomical Head     | 2-Layer MLP + Drop(0.3) | Cross-Entropy Loss               | 7 classes: Arm, Foot, Hand, Lower leg, Thigh, pelvis, wrist |
| Fracture Head       | 2-Layer MLP + Drop(0.3) | BCEWithLogits (pos_weight=2.438) | Binary fracture probability (Threshold >= 0.50)             |

|                   |                                |                           |                                                  |
|-------------------|--------------------------------|---------------------------|--------------------------------------------------|
| Spatial Detector  | Faster R-CNN (MobileNetV3 FPN) | RPN + RoI Box Regression  | Bounding boxes [x1, y1, x2, y2] (Thresh >= 0.10) |
| Caption Generator | Template Synthesizer           | Deterministic rule engine | Factual natural language description             |

#### 4. End-to-End Dashboard Interaction Workflow

- Step 1: Upload** — Doctor or clinical user uploads a plain radiograph into the Streamlit dashboard.
- Step 2: Analysis Trigger** — The 'Analyze X-ray' button dispatches an asynchronous multipart request to FastAPI POST /predict.
- Step 3: Multi-Task Classification** — ResNet-50 predicts the anatomical site and fracture probability.
- Step 4: Spatial Localization** — If fracture is positive, Faster R-CNN detects lesion bounds and generates a visual diagnostic overlay.
- Step 5: Visual Presentation** — Dashboard renders three KPI cards, factual caption, bounding box overlay, and coordinate tables.